

```
In [16]: #Import Libraries
import pandas as pd
import seaborn as sns
import numpy as np

import matplotlib
import matplotlib.pyplot as plt
plt.style.use('ggplot')
from matplotlib.pyplot import figure

%matplotlib inline
matplotlib.rcParams['figure.figsize'] = (12,8) # Adjusts the configuration of the p

# Read in the data

df = pd.read_csv(r'C:\Users\adity\Desktop\PythonJupiter Projects\movies.csv\movies.
```

```
In [17]: # Let's Look at the data
df.head()
```

```
Out[17]:
```

	name	rating	genre	year	released	score	votes	director	writer
0	The Shining	R	Drama	1980	June 13, 1980 (United States)	8.4	927000.0	Stanley Kubrick	Stephen King
1	The Blue Lagoon	R	Adventure	1980	July 2, 1980 (United States)	5.8	65000.0	Randal Kleiser	Henry De Vere Stacpoole
2	Star Wars: Episode V - The Empire Strikes Back	PG	Action	1980	June 20, 1980 (United States)	8.7	1200000.0	Irvin Kershner	Leigh Brackett
3	Airplane!	PG	Comedy	1980	July 2, 1980 (United States)	7.7	221000.0	Jim Abrahams	Jim Abrahams
4	Caddyshack	R	Comedy	1980	July 25, 1980 (United States)	7.3	108000.0	Harold Ramis	Brian Doyle-Murray

```
In [18]: # Lets see if there is any missing data

for col in df.columns:
    pct_missing = np.mean(df[col].isnull())
    print('{} = {}'.format(col, pct_missing))
```

```
name = 0.0%
rating = 0.010041731872717789%
genre = 0.0%
year = 0.0%
released = 0.0002608242044861763%
score = 0.0003912363067292645%
votes = 0.0003912363067292645%
director = 0.0%
writer = 0.0003912363067292645%
star = 0.00013041210224308815%
country = 0.0003912363067292645%
budget = 0.2831246739697444%
gross = 0.02464788732394366%
company = 0.002217005738132499%
runtime = 0.0005216484089723526%
```

```
In [14]: # Data types for our columns
df.dtypes
```

```
Out[14]: name          object
rating         object
genre          object
year           int64
released       object
score          float64
votes          float64
director       object
writer         object
star           object
country        object
budget         float64
gross          float64
company        object
runtime        float64
dtype: object
```

```
In [19]: # change data type of columns
# First fill NaN values with 0 (or another appropriate value) before converting to
df['budget'] = df['budget'].fillna(0).astype('int64')

df['gross'] = df['gross'].fillna(0).astype('int64')
df['votes'] = df['votes'].fillna(0).astype('int64')
```

```
In [20]: df
```

Out[20]:

	name	rating	genre	year	released	score	votes	director	writ
0	The Shining	R	Drama	1980	June 13, 1980 (United States)	8.4	927000	Stanley Kubrick	Steph Kir
1	The Blue Lagoon	R	Adventure	1980	July 2, 1980 (United States)	5.8	65000	Randal Kleiser	Henry I Ve Stacpoc
2	Star Wars: Episode V - The Empire Strikes Back	PG	Action	1980	June 20, 1980 (United States)	8.7	1200000	Irvin Kershner	Leig Bracke
3	Airplane!	PG	Comedy	1980	July 2, 1980 (United States)	7.7	221000	Jim Abrahams	Ji Abrahar
4	Caddyshack	R	Comedy	1980	July 25, 1980 (United States)	7.3	108000	Harold Ramis	Bri Doyl Murr
...	...	...	...	...	...	...	...	...	...
7663	More to Life	NaN	Drama	2020	October 23, 2020 (United States)	3.1	18	Joseph Ebanks	Josep Eban
7664	Dream Round	NaN	Comedy	2020	February 7, 2020 (United States)	4.7	36	Dusty Dukatz	Li Hustc
7665	Saving Mbango	NaN	Drama	2020	April 27, 2020 (Cameroon)	5.7	29	Nkanya Nkwai	Lynr Love
7666	It's Just Us	NaN	Drama	2020	October 1, 2020 (United States)	NaN	0	James Randall	Jam Rand
7667	Tee em el	NaN	Horror	2020	August 19, 2020 (United States)	5.7	7	Pereko Mosia	Perel Mos

7668 rows × 15 columns



```
In [24]: # create correct year column
df['yearcorrect'] = df['released'].str.split(',').str[1].str[:5].str.strip()
```

```
In [53]: df = df.sort_values(by=['score'], inplace=False , ascending=False)
```

```
In [36]: pd.set_option('display.max_rows', 10)
```

```
In [37]: df
```

Out[37]:

	name	rating	genre	year	released	score	votes	director	writ
0	The Shining	R	Drama	1980	June 13, 1980 (United States)	8.4	927000	Stanley Kubrick	Steph Kir
1	The Blue Lagoon	R	Adventure	1980	July 2, 1980 (United States)	5.8	65000	Randal Kleiser	Henry I Ve Stacpoc
2	Star Wars: Episode V - The Empire Strikes Back	PG	Action	1980	June 20, 1980 (United States)	8.7	1200000	Irvin Kershner	Leig Bracke
3	Airplane!	PG	Comedy	1980	July 2, 1980 (United States)	7.7	221000	Jim Abrahams	Ji Abrahar
4	Caddyshack	R	Comedy	1980	July 25, 1980 (United States)	7.3	108000	Harold Ramis	Bri Doyl Murr
...	...	...	...	...	...	...	...	...	...
7663	More to Life	NaN	Drama	2020	October 23, 2020 (United States)	3.1	18	Joseph Ebanks	Josep Eban
7664	Dream Round	NaN	Comedy	2020	February 7, 2020 (United States)	4.7	36	Dusty Dukatz	Li Hustc
7665	Saving Mbango	NaN	Drama	2020	April 27, 2020 (Cameroon)	5.7	29	Nkanya Nkwai	Lynr Love
7666	It's Just Us	NaN	Drama	2020	October 1, 2020 (United States)	NaN	0	James Randall	Jam Rand
7667	Tee em el	NaN	Horror	2020	August 19, 2020 (United States)	5.7	7	Pereko Mosia	Perel Mos

7668 rows × 16 columns



In [12]: # Drop any Duplicates

```
df['company'].drop_duplicates().sort_values(ascending=False)
```

```
Out[12]: 7129    2384
         5664    2383
         6412    2382
         4007    2381
         6793    2380
         ...
         3748     3
         3024     2
         7525     1
         4345     0
         408     -1
         Name: company, Length: 2386, dtype: int16
```

```
In [13]: df.drop_duplicates()
```

```
Out[13]:
```

	name	rating	genre	year	released	score	votes	director	writer	star	coun
0	6587	6	6	1980	1705	8.4	927000.0	2589	4014	1047	
1	5573	6	1	1980	1492	5.8	65000.0	2269	1632	327	
2	5142	4	0	1980	1771	8.7	1200000.0	1111	2567	1745	
3	286	4	4	1980	1492	7.7	221000.0	1301	2000	2246	
4	1027	6	4	1980	1543	7.3	108000.0	1054	521	410	
...	...	...	...	...	...	...	...	...	...	...	...
7663	3705	-1	6	2020	2964	3.1	18.0	1500	2289	2421	
7664	1678	-1	4	2020	1107	4.7	36.0	774	2614	1886	
7665	4717	-1	6	2020	193	5.7	29.0	2061	2683	2040	
7666	2843	-1	6	2020	2817	NaN	NaN	1184	1824	450	
7667	5394	-1	10	2020	391	5.7	7.0	2165	3344	2463	

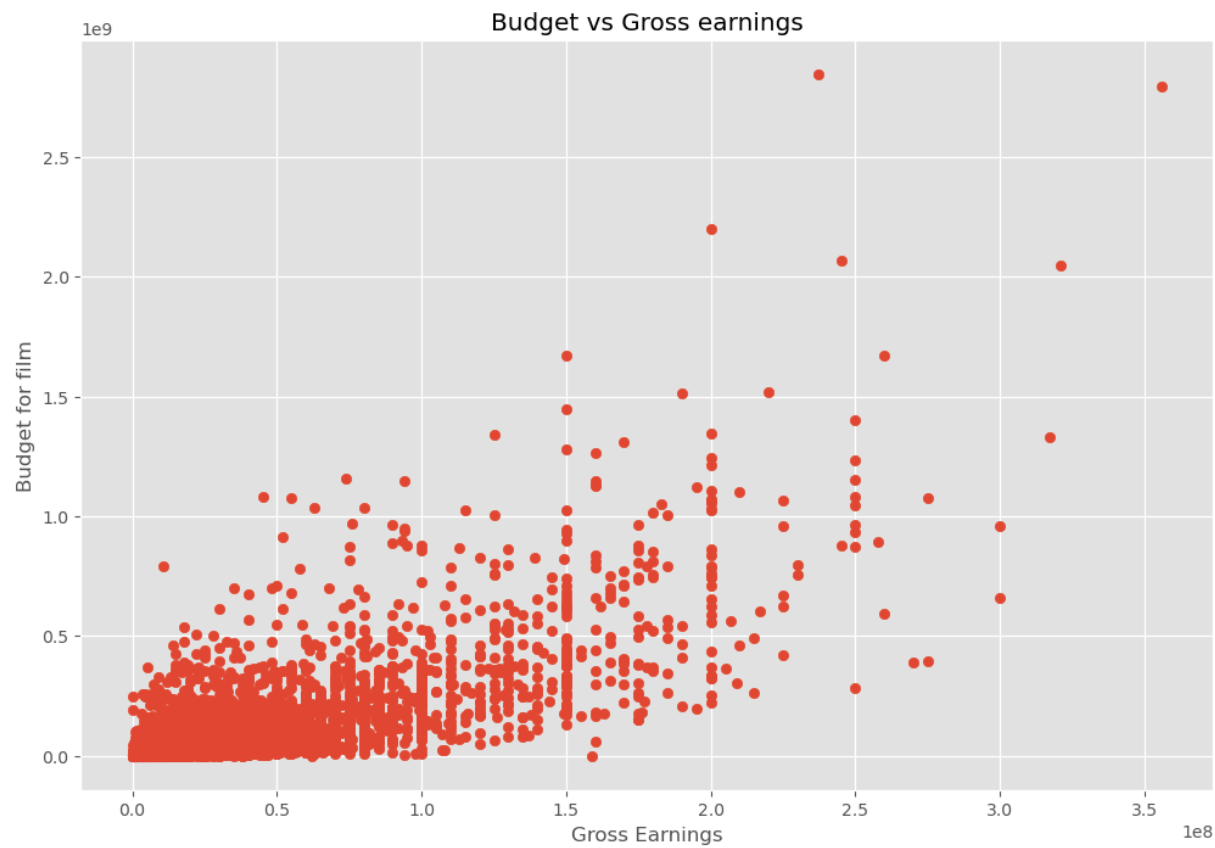
7668 rows × 15 columns



```
In [43]: #budget High Correlation
         # Company high correlation
```

```
In [14]: #scatter plot with budget vs gross

plt.scatter(x=df['budget'], y=df['gross'])
plt.title('Budget vs Gross earnings')
plt.xlabel('Gross Earnings')
plt.ylabel('Budget for film')
plt.show()
```

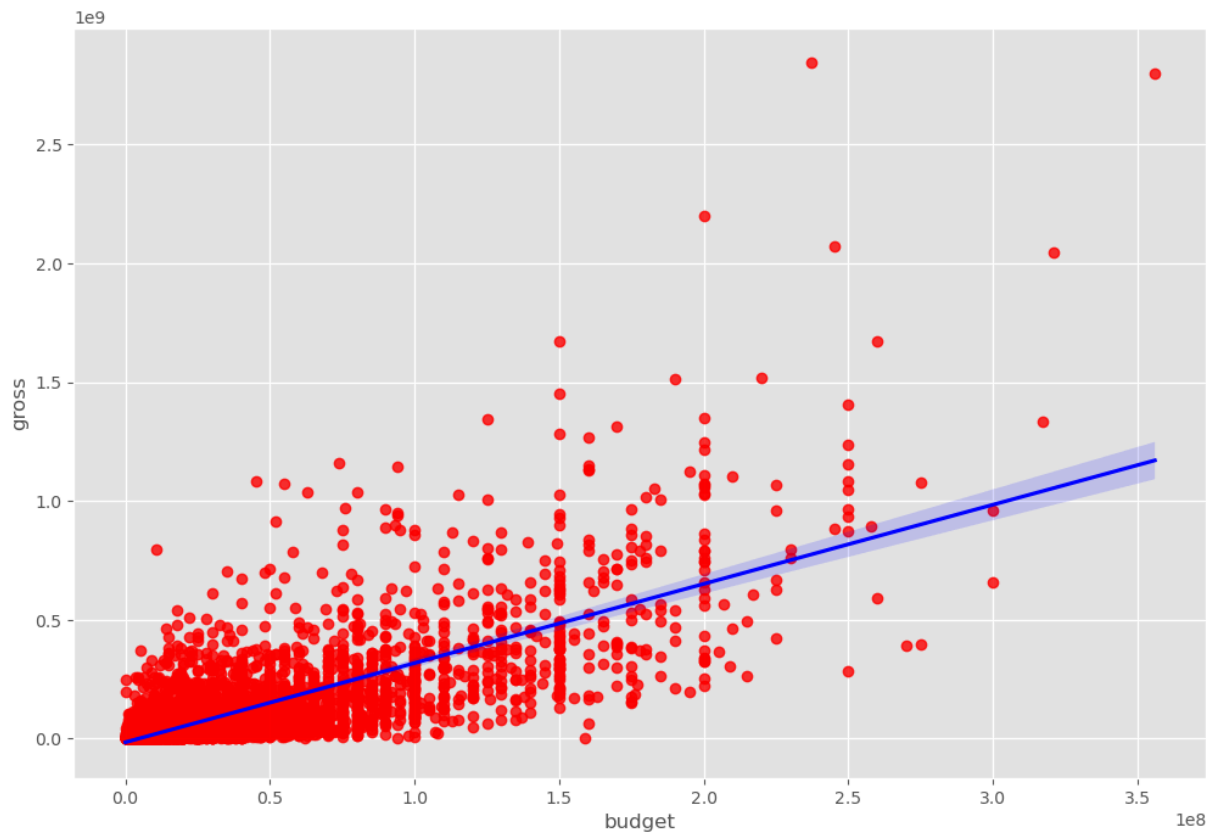


```
In [15]: df.head()
```

Out[15]:

	name	rating	genre	year	released	score	votes	director	writer	star	country
0	6587	6	6	1980	1705	8.4	927000.0	2589	4014	1047	54
1	5573	6	1	1980	1492	5.8	65000.0	2269	1632	327	55
2	5142	4	0	1980	1771	8.7	1200000.0	1111	2567	1745	55
3	286	4	4	1980	1492	7.7	221000.0	1301	2000	2246	55
4	1027	6	4	1980	1543	7.3	108000.0	1054	521	410	55

```
In [9]: # Plot budget vs gross using seaborn
sns.regplot(x='budget', y='gross', data=df , scatter_kws={"color": "red"}, line_kws=
plt.show()
```



In [60]: *#Lets start looking at correlation*

In []:

```
# Select only the numerical columns from the DataFrame
numeric_df = df.select_dtypes(include=['int64', 'float64'])

# Calculate correlation on numeric columns only
correlation_matrix = numeric_df.corr()

# Display the correlation matrix
correlation_matrix
```

Out[8]:

	year	score	votes	budget	gross	runtime
year	1.000000	0.097995	0.222945	0.329321	0.257486	0.120811
score	0.097995	1.000000	0.409182	0.076254	0.186258	0.399451
votes	0.222945	0.409182	1.000000	0.442429	0.630757	0.309212
budget	0.329321	0.076254	0.442429	1.000000	0.740395	0.320447
gross	0.257486	0.186258	0.630757	0.740395	1.000000	0.245216
runtime	0.120811	0.399451	0.309212	0.320447	0.245216	1.000000

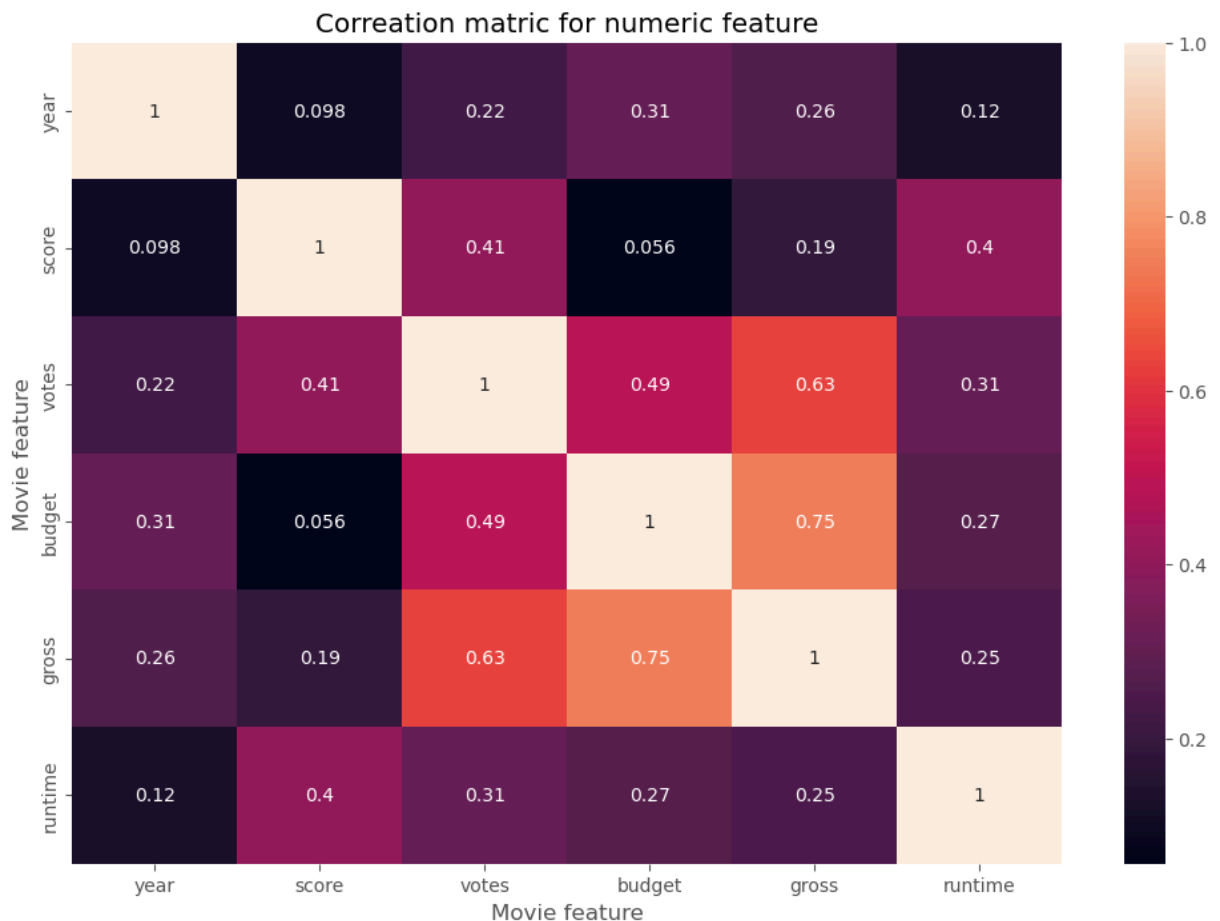
In [68]: *# high Correlation between budget and gross*
I was right

```
In [78]: # Select only the numerical columns from the DataFrame
numeric_df = df.select_dtypes(include=['int64', 'float64'])

# Calculate correlation on numeric columns only
correlation_matrix = numeric_df.corr()

# Display the correlation matrix
correlation_matrix

sns.heatmap(correlation_matrix, annot=True)
plt.title('Correation matric for numeric feature')
plt.xlabel('Movie feature')
plt.ylabel('Movie feature')
plt.show()
```



```
In [5]: # Look at company
df.head()
```

Out[5]:

	name	rating	genre	year	released	score	votes	director	writer
0	The Shining	R	Drama	1980	June 13, 1980 (United States)	8.4	927000.0	Stanley Kubrick	Stephen King
1	The Blue Lagoon	R	Adventure	1980	July 2, 1980 (United States)	5.8	65000.0	Randal Kleiser	Henry De Vere Stacpoole
2	Star Wars: Episode V - The Empire Strikes Back	PG	Action	1980	June 20, 1980 (United States)	8.7	1200000.0	Irvin Kershner	Leigh Brackett
3	Airplane!	PG	Comedy	1980	July 2, 1980 (United States)	7.7	221000.0	Jim Abrahams	Jim Abrahams
4	Caddyshack	R	Comedy	1980	July 25, 1980 (United States)	7.3	108000.0	Harold Ramis	Brian Doyle-Murray

In [6]:

```
df_numerized = df

for col_name in df_numerized.columns:
    if(df_numerized[col_name].dtype == 'object'):
        df_numerized[col_name] = df_numerized[col_name].astype('category')
        df_numerized[col_name] = df_numerized[col_name].cat.codes

df_numerized
```

Out[6]:

	name	rating	genre	year	released	score	votes	director	writer	star	coun
0	6587	6	6	1980	1705	8.4	927000.0	2589	4014	1047	
1	5573	6	1	1980	1492	5.8	65000.0	2269	1632	327	
2	5142	4	0	1980	1771	8.7	1200000.0	1111	2567	1745	
3	286	4	4	1980	1492	7.7	221000.0	1301	2000	2246	
4	1027	6	4	1980	1543	7.3	108000.0	1054	521	410	
...	...	...	...	...	...	...	...	...	...	...	
7663	3705	-1	6	2020	2964	3.1	18.0	1500	2289	2421	
7664	1678	-1	4	2020	1107	4.7	36.0	774	2614	1886	
7665	4717	-1	6	2020	193	5.7	29.0	2061	2683	2040	
7666	2843	-1	6	2020	2817	NaN	NaN	1184	1824	450	
7667	5394	-1	10	2020	391	5.7	7.0	2165	3344	2463	

7668 rows × 15 columns



In [11]:

```
df
```

Out[11]:

	name	rating	genre	year	released	score	votes	director	writer	star	coun
0	6587	6	6	1980	1705	8.4	927000.0	2589	4014	1047	
1	5573	6	1	1980	1492	5.8	65000.0	2269	1632	327	
2	5142	4	0	1980	1771	8.7	1200000.0	1111	2567	1745	
3	286	4	4	1980	1492	7.7	221000.0	1301	2000	2246	
4	1027	6	4	1980	1543	7.3	108000.0	1054	521	410	
...	...	...	...	...	...	...	...	...	...	...	
7663	3705	-1	6	2020	2964	3.1	18.0	1500	2289	2421	
7664	1678	-1	4	2020	1107	4.7	36.0	774	2614	1886	
7665	4717	-1	6	2020	193	5.7	29.0	2061	2683	2040	
7666	2843	-1	6	2020	2817	NaN	NaN	1184	1824	450	
7667	5394	-1	10	2020	391	5.7	7.0	2165	3344	2463	

7668 rows × 15 columns



In [20]:

```
df
```

Out[20]:

	name	rating	genre	year	released	score	votes	director	writ
0	The Shining	R	Drama	1980	June 13, 1980 (United States)	8.4	927000	Stanley Kubrick	Steph Kir
1	The Blue Lagoon	R	Adventure	1980	July 2, 1980 (United States)	5.8	65000	Randal Kleiser	Henry I Ve Stacpoc
2	Star Wars: Episode V - The Empire Strikes Back	PG	Action	1980	June 20, 1980 (United States)	8.7	1200000	Irvin Kershner	Leig Bracke
3	Airplane!	PG	Comedy	1980	July 2, 1980 (United States)	7.7	221000	Jim Abrahams	Ji Abrahar
4	Caddyshack	R	Comedy	1980	July 25, 1980 (United States)	7.3	108000	Harold Ramis	Bri Doyl Murr
...	...	...	...	...	...	...	...	...	...
7663	More to Life	NaN	Drama	2020	October 23, 2020 (United States)	3.1	18	Joseph Ebanks	Josep Eban
7664	Dream Round	NaN	Comedy	2020	February 7, 2020 (United States)	4.7	36	Dusty Dukatz	Li Hustc
7665	Saving Mbango	NaN	Drama	2020	April 27, 2020 (Cameroon)	5.7	29	Nkanya Nkwai	Lynr Love
7666	It's Just Us	NaN	Drama	2020	October 1, 2020 (United States)	NaN	0	James Randall	Jam Rand
7667	Tee em el	NaN	Horror	2020	August 19, 2020 (United States)	5.7	7	Pereko Mosia	Perel Mos

7668 rows × 15 columns

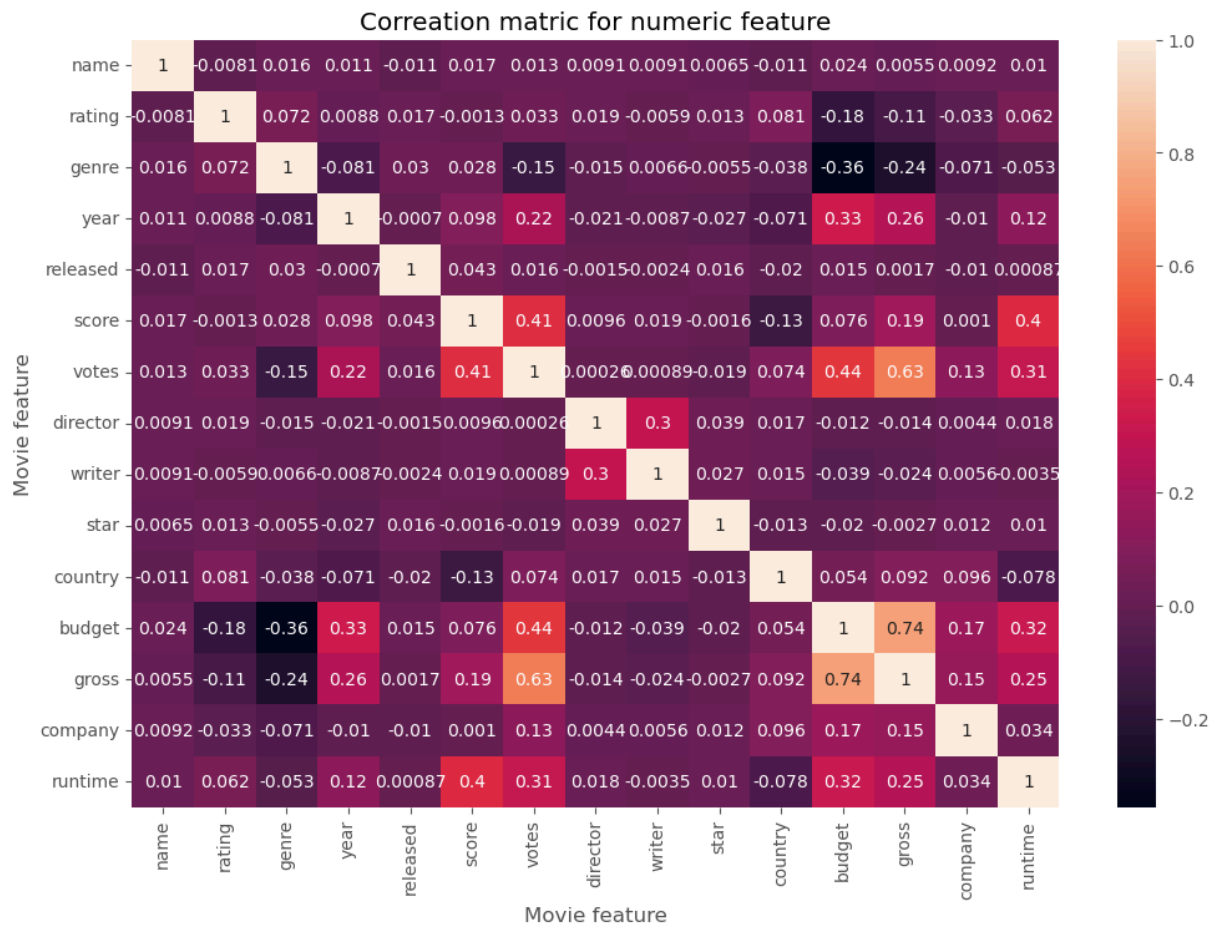


```
In [21]: # Select only the numerical columns from the DataFrame
numeric_df = df.select_dtypes(include=['int64', 'float64'])
```

```
# Calculate correlation on numeric columns only
correlation_matrix = df_numerized.corr()

# Display the correlation matrix
correlation_matrix

sns.heatmap(correlation_matrix, annot=True)
plt.title('Correation matrix for numeric feature')
plt.xlabel('Movie feature')
plt.ylabel('Movie feature')
plt.show()
```



In [22]: `df_numerized.corr()`

Out[22]:

	name	rating	genre	year	released	score	votes	dire
name	1.000000	-0.008069	0.016355	0.011453	-0.011311	0.017097	0.013088	0.009
rating	-0.008069	1.000000	0.072423	0.008779	0.016613	-0.001314	0.033225	0.019
genre	0.016355	0.072423	1.000000	-0.081261	0.029822	0.027965	-0.145307	-0.019
year	0.011453	0.008779	-0.081261	1.000000	-0.000695	0.097995	0.222945	-0.020
released	-0.011311	0.016613	0.029822	-0.000695	1.000000	0.042788	0.016097	-0.001
score	0.017097	-0.001314	0.027965	0.097995	0.042788	1.000000	0.409182	0.009
votes	0.013088	0.033225	-0.145307	0.222945	0.016097	0.409182	1.000000	0.000
director	0.009079	0.019483	-0.015258	-0.020795	-0.001478	0.009559	0.000260	1.000
writer	0.009081	-0.005921	0.006567	-0.008656	-0.002404	0.019416	0.000892	0.299
star	0.006472	0.013405	-0.005477	-0.027242	0.015777	-0.001609	-0.019282	0.039
country	-0.010737	0.081244	-0.037615	-0.070938	-0.020427	-0.133348	0.073625	0.017
budget	0.023970	-0.176002	-0.356564	0.329321	0.014683	0.076254	0.442429	-0.012
gross	0.005533	-0.107339	-0.235650	0.257486	0.001659	0.186258	0.630757	-0.014
company	0.009211	-0.032943	-0.071067	-0.010431	-0.010474	0.001030	0.133204	0.004
runtime	0.010392	0.062145	-0.052711	0.120811	0.000868	0.399451	0.309212	0.017



```
In [23]: correlation_mat = df_numerized.corr()
corr_pairs = correlation_mat.unstack()
corr_pairs
```

```
Out[23]: name      name      1.000000
          rating    -0.008069
          genre      0.016355
          year       0.011453
          released  -0.011311
          ...
runtime country  -0.078412
          budget   0.320447
          gross     0.245216
          company   0.034402
          runtime   1.000000
Length: 225, dtype: float64
```

```
In [24]: sorted_pairs = corr_pairs.sort_values()
sorted_pairs
```

```
Out[24]: budget    genre    -0.356564
         genre    budget    -0.356564
         gross    -0.235650
         gross    genre    -0.235650
         rating   budget    -0.176002
         ...
         year     year      1.000000
         genre    genre      1.000000
         rating   rating      1.000000
         company  company     1.000000
         runtime  runtime     1.000000
         Length: 225, dtype: float64
```

```
In [26]: high_corr = sorted_pairs[(sorted_pairs) > 0.5]
         high_corr
```

```
Out[26]: gross      votes      0.630757
         votes      gross      0.630757
         budget     gross      0.740395
         gross      budget     0.740395
         name       name       1.000000
         director   director   1.000000
         gross      gross      1.000000
         budget     budget     1.000000
         country    country    1.000000
         star       star       1.000000
         writer     writer     1.000000
         votes      votes      1.000000
         score      score      1.000000
         released   released   1.000000
         year       year       1.000000
         genre      genre      1.000000
         rating     rating     1.000000
         company    company    1.000000
         runtime    runtime    1.000000
         dtype: float64
```

```
In [27]: #votes and budget have highest correlation to gross earnings
         # company has low correlation
```

```
In [ ]:
```